
MODEL SELECTION AND VALIDATION REPORT (remove_8_biomass model)

OBSERVATIONS:1687

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_8_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.31	0.17
remove_8_kde_fric	-0.02	0.04
remove_8_kde_feve	-0.29	0.03
sbt_mean	0.44	0.10
sbt_sd	0.02	0.05
sst_sd	0.01	0.05
log_chla_mean	0.08	0.04
depth	0.03	0.05
log_sum_fishing	0.06	0.04

Dispersion parameter: 1.05

Spatiotemporal AR1 correlation (rho): 0.91

Matérn range: 190.66

Spatiotemporal marginal AR1 SD: 0.91

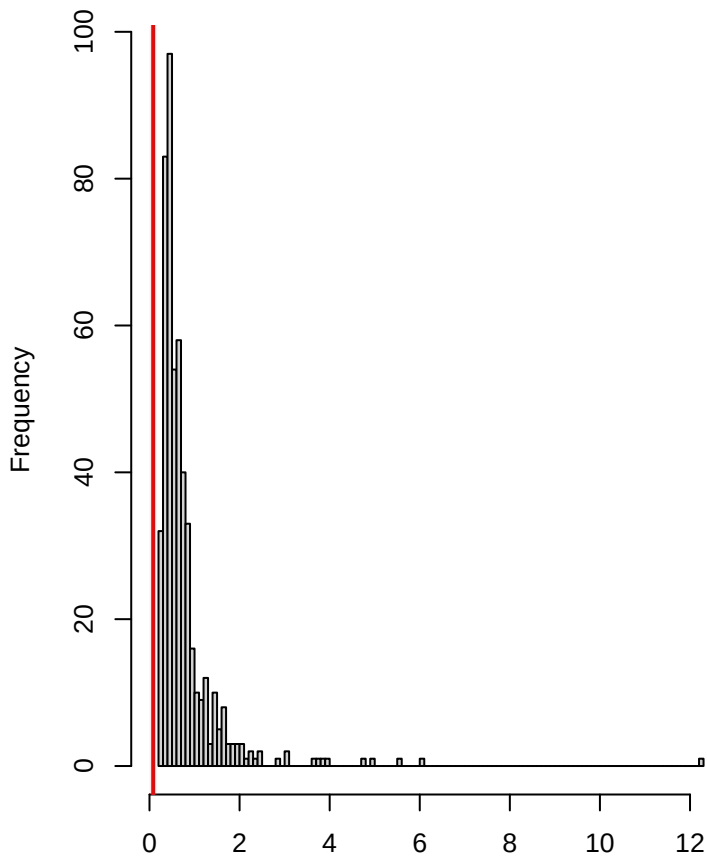
REML criterion at convergence: 10588.700

See ?tidy.sdmTMB to extract these values as a data frame.

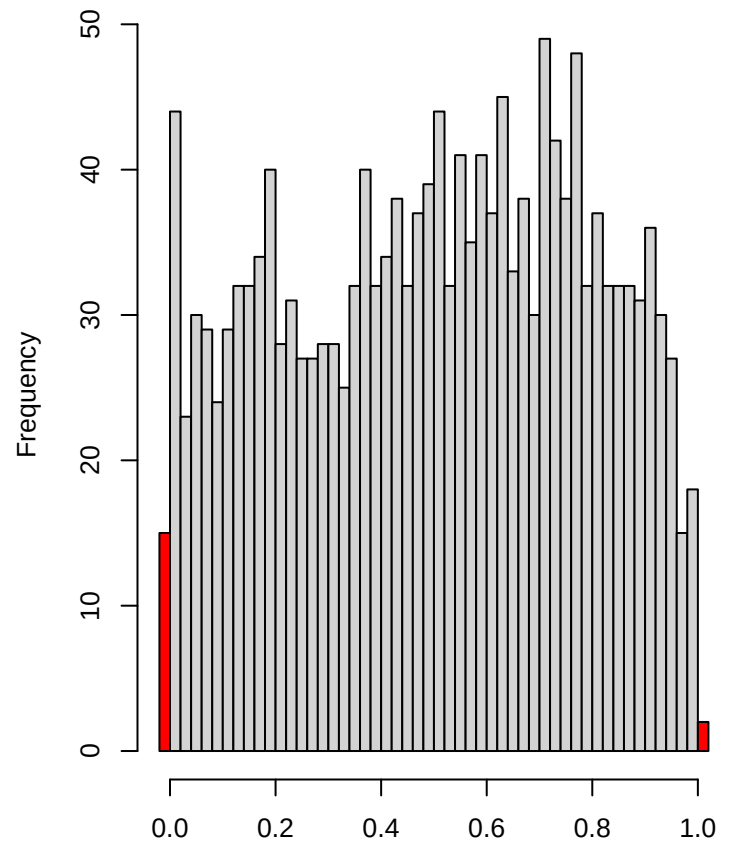
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	21203.40	0.9584302
spatiotemporal_random_fields_ar1_with_anisotropy	15	21204.36	75.0800159
spatial_random_fields	12	21279.44	3.7013910
spatial_random_fields_with_anisotropy	14	21283.14	106.5481632
spatiotemporal_random_fields_iid_with_anisotropy	14	21389.69	1.2305857
spatiotemporal_random_fields_iid	12	21390.92	136.3672104
error_only	10	21527.29	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



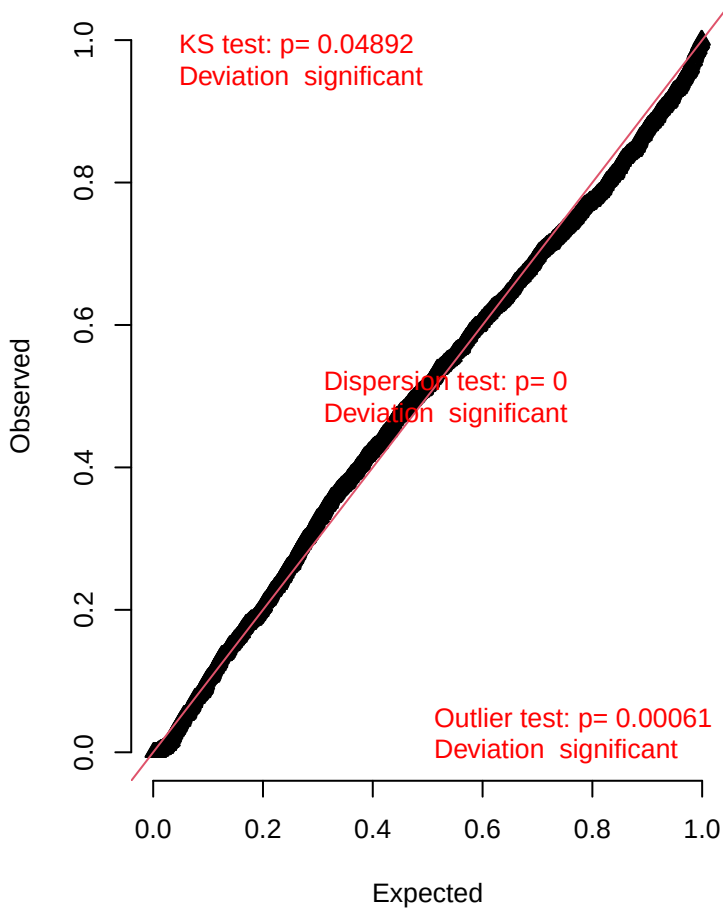
Outlier test significant



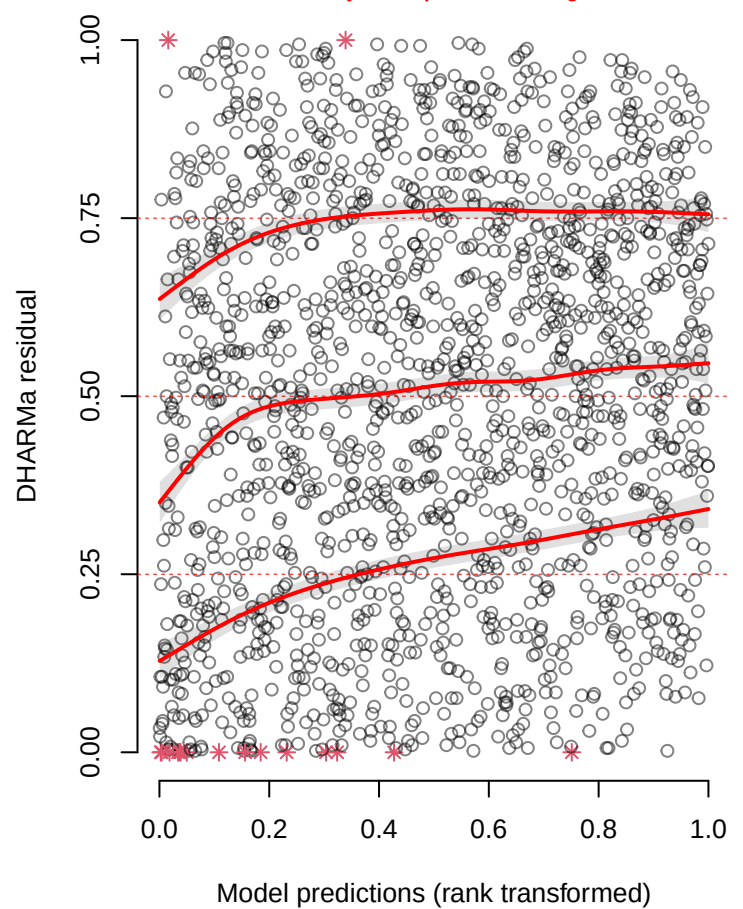
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.023 | H1: Distance-based autocorrelation | p-value = 0.012

